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import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("/content/titanic - titanic.csv")

# -------------------------------
# 1 BASIC STATISTICAL ANALYSIS
# -------------------------------

print("Dataset Info:")
print(df.info())

print("\nStatistical Summary:")
print(df.describe())

print("\nMissing Values Count:")
print(df.isnull().sum())

# -------------------------------
# 2 DATA VISUALIZATION
# -------------------------------

# Histogram - Age
plt.hist(df['Age'].dropna(), bins=10)
plt.xlabel("Age")
plt.ylabel("Count")
plt.title("Age Distribution")
plt.show()

# Histogram - Fare
plt.hist(df['Fare'].dropna(), bins=10)
plt.xlabel("Fare")
plt.ylabel("Count")
plt.title("Fare Distribution")
plt.show()

# Survival count bar plot
df['Survived'].value_counts().plot(kind='bar')
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of Passengers")
plt.title("Survival Count")
plt.show()

# Boxplot for Age and Fare
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```python
df[['Age', 'Fare']].plot(kind='box')
plt.title("Boxplot for Age and Fare")
plt.show()

# -------------------------------
# 3 OUTLIER ANALYSIS (IQR METHOD)
# -------------------------------

Q1 = df['Fare'].quantile(0.25)
Q3 = df['Fare'].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[(df['Fare'] < lower_bound) | (df['Fare'] > upper_bound)]

print("\nFare Outliers:")
print(outliers[['Fare']].head())

print("\nNumber of Fare Outliers:", len(outliers))
```